

# MUHAMAD SOLEHUDDIN

## MANUFACTURING AUTOMATION ENGINEER

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### PROFESSIONAL PROFILE

Manufacturing automation engineer and Industrial Automation and Robotics engineering technologist with experience spanning production-equipment maintenance, factory digitization, industrial communications, engineering software, and practical troubleshooting. Connects physical equipment to reliable information systems using Modbus, RS-485, MQTT, Python, SQL databases, dashboards, and documented operating workflows. Builds evidence-led demonstrations in MES, machine vision, and robotics while clearly separating deployed work, prototypes, engineering concepts, and work in progress.

### CORE ENGINEERING CAPABILITIES

- **Manufacturing automation:** process observation, bottleneck framing, equipment telemetry, PLC and HMI foundations, interlocks, sequencing, and automation trade studies
- **Industrial integration:** Modbus RTU/TCP, RS-485, MQTT, edge Linux devices, Node-RED, Grafana, HMI/SCADA concepts, and OT-to-IT data paths
- **Software and data:** Python, FastAPI, REST APIs, SQL, PostgreSQL, TimescaleDB, Docker Compose, Git, automated tests, dashboards, and technical runbooks
- **Equipment and fabrication:** structured fault isolation, preventive maintenance, electromechanical assembly foundations, SolidWorks, Mastercam, CNC programming, basic industrial wiring, milling, and lathe
- **Emerging engineering areas:** ROS 2 interfaces, computer vision, discrete-event process simulation, AMR/ASRS concepts, MES/ERP workflows, and traceability

### PROFESSIONAL EXPERIENCE

#### Freelance / Project-based System Engineer

Independent Engineering Projects | Penang, Malaysia | Oct 2025-Aug 2026

- Built and documented self-directed industrial learning systems covering machine telemetry, APIs, databases, dashboards, PLC/HMI concepts, manufacturing information flow, and local deployment.
- Developed the NEXUS IIoT validated demonstration using Python, FastAPI, MQTT, TimescaleDB, Svelte, Grafana, Docker Compose, automated checks, and documented operating scenarios.
- Built a sanitized MES/ERP manufacturing-flow demonstrator covering synthetic orders, material state, work execution, traceability, quality events, and reporting.
- Produced architecture notes, setup instructions, test evidence, limitations, and troubleshooting runbooks so each project can be reviewed and reproduced with its maturity stated explicitly.

#### System Engineer, Digitization Department

Alumac Industries Sdn. Bhd. | Sungai Buloh, Malaysia | Jan 2024-Sep 2025

- Developed and supported a real-time factory energy-visibility system across multiple production assets using industrial meters, Modbus RTU over RS-485, Python edge services, SQL storage, and Node-RED dashboards.
- Diagnosed faults across physical signals, serial communications, edge services, databases, and operational dashboards in a live manufacturing environment.
- Integrated a PLC-controlled legacy production asset into an IIoT monitoring path to improve operating visibility and maintenance troubleshooting.
- Translated production, maintenance, management, and IT needs into monitoring logic, decision-facing views, documentation, and handover steps.
- Supported manufacturing reporting and Microsoft 365 workflows using Power BI, SharePoint, Teams, Power Automate, and Power Apps.

## Maintenance Technician

Diversatech (M) Sdn. Bhd. | Klang, Malaysia | May 2021-Dec 2023

- Performed diagnostics, preventive maintenance, machine repair, and structured troubleshooting on production equipment.
- Supported engineering teams with fault isolation and recovery while recording machine condition, observed behaviour, and corrective actions for escalation.

## CAD and CAM Technician

AMS Seihin Sdn. Bhd. | Kajang, Malaysia | Jun 2016-Sep 2018

- Programmed and troubleshoot CNC equipment, including Brother and HAAS machining centres, using Mastercam and shop-floor verification.
- Used SolidWorks for design and assembly work and supported industrial wiring, milling, lathe, machine repair, and mechanical troubleshooting.

## SELECTED ENGINEERING PORTFOLIO

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### NEXUS Industrial IIoT - Validated Demonstration

- Connected simulated manufacturing signals through Modbus, MQTT, TimescaleDB, FastAPI, Grafana, and a browser decision view; documented reproducible fault and alert scenarios.

### MES / ERP Manufacturing Flow - Validated Demonstration

- Modelled synthetic order, material, execution, quality, traceability, and reporting flows in a sanitized manufacturing-information system; not presented as a production MES deployment.

### Asset Vision - Prototype

- Built a hardware-independent inspection path using Python, FastAPI, OpenCV, local records, a browser dashboard, typed ROS 2 interfaces, supplied sample images, and automated tests.

### Tube Preparation Automation - Engineering Concept

- Decomposed a generic repetitive preparation process into verification, measurement, identification, guarded separation, inspection, output, and kitting modules for an automation trade study.
- Uses invented or generalized inputs only; unvalidated, unaffiliated with any employer, and makes no production, savings, quality, or throughput claim.

## EDUCATION AND PROFESSIONAL CREDENTIAL

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- **Bachelor of Engineering Technology in Industrial Automation and Robotics Technology** - Universiti Kuala Lumpur Malaysia France Institute (UniKL MFI) | 2017-2023
- **Diploma in Mechatronics** - German-Malaysian Institute
- **Graduate Technologist (GT), Manufacturing and Industrial Technology** - Malaysia Board of Technologists (MBOT) | 2026